Report on "Matrix Theory Reloaded: A BPS Road to Holography"

(Blair, Lahnsteiner, Obers, Yan)

Summary of the paper. The paper revisits the decoupling limits that define matrix theories on Dp-branes ("matrix p-brane theory", MpT). It stresses that these limits are BPS limits, which require tuning the RR potential to the critical value at which the brane tension equals its charge. They make the ten-dimensional target space non-Lorentzian, with a p-brane Newton–Cartan type foliation. Three claims follow. (i) Applied to a curved Dp-brane background, the same asymptotic MpT limit produces the near-horizon limit, so $\mathrm{AdS}_5 \times S^5$ and the IMSY geometries appear as "MpT geometries". (ii) Applying two such limits (Dp on a Dq background, or D1–D5) gives non-Lorentzian holographic duals. Lifting to M-theory, each BPS limit is a DLCQ, which leads to a conjectured "DLCQ^n/DLCQ^m" classification of holography. (iii) Undoing the limit is a p-brane generalisation of the $T\bar{T}$ deformation. The authors derive explicit classical flow equations, summarised below.

- Free scalars flow to the Dirac–Nambu–Goto action for any p.
- For $p = 1, 2$, scalars plus a $U(1)$ gauge field flow to the DBI action.
- For $p = 3$, the flow equation is known only to linear order in the deformation parameter.

1. Principal achievement

The main achievement is a unifying, geometric reading of the relation between matrix theory and AdS/CFT. The Maldacena/IMSY near-horizon limit in the bulk is the same BPS limit as the one that defines the asymptotic matrix-theory sector. The $(r/\ell)^2$ warp factor of AdS_5 plays the role of the limiting parameter ω, so the bulk is a "geometrised" version of the MpT limit. This is not entirely new: the $p = 0$ case was understood in the late 1990s. The paper makes it systematic for all p and shows that it extends to multicritical limits. This yields a concrete recipe for non-Lorentzian near-horizon geometries and recovers AdS_3 from a double limit on D1–D5.

The most concrete and checkable by-product is the set of p-brane $T\bar{T}$ flow equations. Their key ingredient is a λ-dependent term, which explains why the ordinary $T\bar{T}$ deformation of the multi-scalar theory does not reach Nambu–Goto and why $T\bar{T}$ of 1+1d Maxwell does not reach DBI. The $p = 2$ equation, $\partial_\lambda \mathcal{L} = \frac{1}{4}[\mathrm{tr}\,\mathcal{T}^2 - (\mathrm{tr}\,\mathcal{T})^2] + \frac{\lambda}{2}\det \mathcal{T}$, is the same for the M2 and the D2, and I find it elegant.

2. Principal drawback

The paper is largely kinematic, and the central claims lack independent dynamical tests.

- The non-Lorentzian bulk geometries are obtained by rescaling the harmonic-function scale ℓ by hand with ω. The authors state in the Outlook that they have not verified that these geometries solve the equations of motion and torsional constraints of MpT supergravity. They also state that the (super)symmetries, charges and isometries are not analysed, and that the physical meaning of the rescaling of N is unclear. So the "non-Lorentzian

holography" is a family of candidate backgrounds, not established dual pairs, and no dictionary or observable is matched.

- The DLCQ^n/DLCQ^m conjecture is essentially bookkeeping: count the "1"s dropped from harmonic functions and the lightlike circles. It is illustrated on single-stack and D1–D5 examples and not tested beyond them. It is not clear what it predicts that could fail.
- The "bulk from a $T\bar{T}$ deformation" statement amounts to promoting $\omega \to \Omega(r)$ using the emergent dilatation symmetry, with Ω chosen to reproduce the known solution. It is a reinterpretation rather than a derivation, and the proposal to solve for Ω from the equations of motion is left for the future.
- The flow equations are classical, bosonic and abelian. They are closed-form only for $p \leq 2$, and $p = 3$ is known only perturbatively. The cases $p > 3$ are strongly coupled and require a change of duality frame. The non-abelian and supersymmetric cases, which would be the interesting ones for matrix theory and SYM, are open.

3. Most worthwhile follow-up

Proposal: formulate the MpT supergravity equations and use them to test and classify the candidate non-Lorentzian near-horizon geometries and their symmetries.

Concretely:

1. Derive the bosonic equations of motion and torsional constraints of MpT supergravity as the $\omega \to \infty$ limit of type II supergravity, starting with $p = 0, 1$, where the non-relativistic and Newton–Cartan technology is most developed.
2. Insert the paper's ansatz (the $\tilde{\not{<}}$-dependent MpT geometry with a single function $\tilde{\not{<}}(r)$) and the localised and smeared Dq geometries into these equations. This decides which candidates are genuine solutions and which brane-charge and smearing choices are needed.
3. Compute the isometry algebras of the solutions that survive. These include the dilatation and boost-like symmetries and the analogue of conformal or Schrödinger-type algebras. Compare them with the symmetries of the dual non-relativistic SYM or matrix-theory sector.
4. Add a finite-temperature (black-brane) version of the bulk solution.

Why this one. It is feasible. It is a well-defined calculation that the authors themselves identify as needed. It builds on existing non-relativistic supergravity results (for example, the M2-brane analyses in non-relativistic M-theory) and requires no new conceptual machinery. It is also the step that decides whether the paper's central proposals become physics or remain a classification. The symmetry match supplies the first genuine falsifiable check of the DLCQ^n/DLCQ^m picture, since the dual field theory's symmetry is fixed independently by the DLCQ orbit. Thermodynamic data would supply the first observable. The alternatives are less attractive. The non-abelian or supersymmetric extension of the $p = 2$ flow equation (BLG/ABJM or the M2-brane) has higher potential impact, but it is much harder and has no clear criterion for the correct deformation. Higher-p closed-form flows are likely to be algebraically heavy and of narrower interest.